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SOTA: How to explain a classifier?

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Abstract

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Introduction

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1 Perturbation model-agnostic methods

In this section, we present a set of perturbation method used in black-box models explanation. A perturbation method can be defined by the generation of perturbed samples (ex: noised features in model representation or perturbed input tensor) at the input of the black-box model. The objective is to compare the model behavior (the prediction) between a set of different perturbed inputs and the original one. Then, we can determine which one is more important for the prediction. It allows the explanation model to detail why the model took a specific decision and why. The following methods register in this type. The first one, LIME (Local Interpretable Model-Agnostic Explanation) focus on a linear approximation of local model behavior. Secondly, the Anchors method provides an amelioration of LIME by introducing the concept of IF-THEN rules to provide a better user-experience. Finally, we will briefly talk about add-ons of those methods in specific cases.

Remark (Notation). Let's introduce some notations to identify which objects we will talk about. We denote with f a black-box model we want to explain, then $f : X \longrightarrow Y$ where X is the set of instances $x \in X$ and Y the different possible predictions $f(x)$. In this section, we will talk about linear explanation model, denoted g. To identify which input representation we use, we denote by x' the interpretable representation of an input x in the black-box

model f. Then, if we construct a set of perturbed interpreted samples $\mathcal{Z}$ from x', this set contains perturbed version of x' called z'. Finally, $\Omega(g)$ express the complexity of the model g, generally defined by the number of non-zeros weights in g.

1.1 LIME (Local Interpretable Model-Agnostic Explainer)

LIME founded the principe of perturbation method for classifiers explanation [1]. It provides a local explanation of a black-box model behavior $f : \mathbb{R}^d \longrightarrow \mathbb{R}$ by training a simple explanation model g over perturbed samples. This method is *model-agnostic* because it doesn't depend on the black-box model architecture. f is considered as an oracle used to train g.

Formally, for an original interpretable representation $x' \in \{0,1\}^{d'}$ of a sample $x \in \mathbb{R}^d$, it generates a set of perturbed samples z' which are drawn by selecting uniformly some x' non-zero components. This random sampling is weighted by π_x which defines a locality around x' denoted $\mathcal{Z}$. In practical, π_x is an exponential kernel applied to a distance D. To find the optimal model g, we can define a loss function $\mathcal{L}(f, g, \pi_x)$ telling how unfaithful g is in approximating f on $\mathcal{Z}$. The objective is then to minimize $\mathcal{L}$ and having $\Omega(g)$ low enough for human comprehension. The LIME explanation is then produced by:

$$\xi(x) = \arg\min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g) \tag{1}$$

with G the set of available explanation models [1]. Using $\mathcal{L}$ and $\Omega(g)$ to find the best model g ensures *interpretability* and *local fidelity* on the black-box model explanation. Finally, the local explanation of the model is provided by the weights w_g trained with $g(z') = w_g \cdot z'$.

As stated at beginning, LIME provides a *local* explanation of f. To ensure a global explanation, the paper proposes a Sub-modular Picks LIME (SP-LIME) which selects a set local and non-redundant LIME explanations to be given to the user [1]. To choose those explanations, the paper introduces the concept of coverage of a set of instances which defines the importance of those instances.

Definition 1.1 *Coverage*

Let g be an explanation model and $g_i = \xi(x_i)$ the explanation of the instance x_i, we construct the $|X| \times d'$ explanation matrix $\mathcal{W}$ with $\mathcal{W}_{i,j} = |w_{g_{i,j}}|$. Then, I_j is the j-column of W and denotes the global importance of that component in the explanation space. Then, the coverage of a set of instances V is defined as the total importance of the features that appear at least one instance in V:

$$c(V, \mathcal{W}, I) := \sum_{j=1}^{d'} \mathbb{1}_{\{\exists i \in V, \mathcal{W}_{i,j} > 0\}} I_j \tag{2}$$

Finally, from a maximum number of global explanations B set by the user, the pick problem is defined by:

$$Pick(\mathcal{W}, I) = \arg\max_{V, |V| \leqslant B} c(V, \mathcal{W}, I) \tag{3}$$

We want to select the highest coverage and non-redundant explanations to provide a global one.

Unfortunately, this system has limits. First, in practice, the pick problem is intractable, so a greedy algorithm is used to compute it. Secondly, LIME algorithm doesn't define clear boundaries of the explanation instances. Indeed, all black-box models cannot be locally approximated by a linear one [2]. Finally, the explanation results depend on the LIME parameters such as B, the distance metric D and the kernel π_x. Changing one of them can change the interpretation and undermining the reliability of this method [3].

1.2 Anchors

Anchors is also a model-agnostic explanation method based on LIME system but enhanced in order to provide local sufficient conditions for prediction. Anchors are defined by short and disjoint rules which are preferred by users for explanation [4]. It can be applied on different set of domains (tabular, text, images, etc...) for different tasks (classification, prediction, text generation, etc...).

Definition 1.2 *Anchors*

Formally, Anchors are built on a set of predicates a called a "rules". We can also see it as a function of instances $x \in X$:

$$A : \begin{cases} X \longrightarrow \{0,1\} \\ x \longmapsto \begin{cases} 1 \text{ if all rules are true for } x \\ 0 \text{ otherwise} \end{cases} \end{cases}$$

to link a set of rules A on a prediction $f(x)$ (where $f : X \longrightarrow Y$ is the black-box model). Finally, A is an Anchor of x if $A(x) = 1$ and A is a sufficient condition for the prediction $y = f(x)$:

$$\mathbb{E}_{\mathcal{D}(z|A)}[1_{f(x)=f(z)}] \geqslant \tau \tag{4}$$

with τ the desired level of precision of the Anchor on x [4]. Here, $\mathcal{D}(x \mid A)$ represents the distribution of neighbors of x which satisfy the Anchor A.

Computing Anchors is the difficult part of the process. This is a combinatorial optimization problem if we decide to choose anchors which satisfy a precision constraint with high probability:

$$\mathbb{P}(prec(A) \geqslant \tau) \geqslant 1 - \delta$$

with $prec(A) = \mathbb{E}_{\mathcal{D}(z|A)}(1_{f(x)=f(z)})$ called the Anchors "precision". Among those anchors, we choose the largest coverage ones, where coverage is :

$$cov(A) = \mathbb{E}_{\mathcal{D}(z)}(A(z)) \tag{5}$$

Finally, the optimization problem can be described by :

$$\max_{\mathbb{P}(prec(A)\geqslant\tau)\geqslant 1-\delta} cov(A) \tag{6}$$

Because the number of all possible Anchor is exponential, we can perform an exhaustive search. That's why the article introduced the Bottom-Up approach:

1. *Initialization*: The anchor A is initialized as an empty rule (applying to all instances).
2. *Candidate Generation*: In each iteration, a set of candidate rules $\mathcal{A}$ is generated by extending A with one additional feature predicate $\{a_i\}$: $\mathcal{A} = \{A \wedge a_i, A \wedge a_{i+1}, A \wedge a_{i+2}, ...\}$
3. *Selection via Multi-Armed Bandit (MAB)*: To identify the best candidate without an exhaustive number of calls to f, the problem is formulated as a pure-exploration multi-armed bandit problem. The *KL-LUCB* algorithm is used to find the rule with the highest precision using a minimal set of samples [4].
4. *Termination*: The process terminates when a rule meets the statistical precision constraint: $\mathbb{P}(prec(A) \geqslant \tau) \geqslant 1 - \delta$.

To address the limitations of a purely greedy search, the algorithm extends the construction to a *Beam Search* of size B. It maintains a set of B current candidates instead of a single one. Among the multiple anchors identified that satisfy the precision threshold τ, the algorithm outputs the one with the highest coverage $cov(A)$, effectively optimizing the original problem.

To provide a global understanding of the model, the article adapts a *Submodular Pick* (SP) procedure. The goal is to select a set of K anchors that are non-redundant and cover as many instances in the validation set similar to the approach used in LIME [1, 4].

References

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